

9.3.2 Mutual Information

$X \backslash Y$	0	1
0	0.25	0.15
1	0.1	0.5

Q: $MI(X, Y) = ?$

Solution: ① $p(X=0) = 0.25 + 0.15 = 0.4$

$p(X=1) = 0.1 + 0.5 = 0.6$

$p(Y=0) = 0.25 + 0.1 = 0.35$

$p(Y=1) = 0.15 + 0.5 = 0.65$

② $E(X) = -p(X=0) \log_2 p(X=0) - p(X=1) \log_2 p(X=1)$

$= -0.4 \times \log_2 0.4 - 0.6 \log_2 0.6$

$= 0.9710$

③ $E(Y) = -p(Y=0) \log_2 p(Y=0) - p(Y=1) \log_2 p(Y=1)$

$= -0.35 \log_2 0.35 - 0.65 \log_2 0.65$

$= 0.9341$

④ $E(X, Y) = -p(X=0, Y=0) \log_2 (p(X=0, Y=0))$

$$-p(X=0, Y=1) \log_2(p_{X=0, Y=1})$$

$$-p(X=1, Y=0) \log_2(p_{X=1, Y=0})$$

$$-p(X=1, Y=1) \log_2(p_{X=1, Y=1})$$

$$= -0.25 \log_2 0.25 - 0.15 \log_2 0.15 - 0.1 \log_2 0.1 - 0.5 \log_2 0.5$$

$$= 1.7427$$

$$MI(X, Y) = E(X) + E(Y) - E(X, Y)$$

$$= 0.9710 + 0.9341 - 1.7427$$

$$= 0.1624$$